Study Questions for Lectures on Inheritance KEY

1. Answer is B
2. Answer is C
3. Answer is C
4. Answer is C
5. Answer is A and B
6.
    a. Methionine (AUG) and Tryptophan (UGG). *If you also listed stop codons, note that there are three stop codons (UGA, UAA, UAG)*
    b. All of them, except the two above: Phenylalanine, Leucine, Isoleucine, Valine, Serine, Proline, Threonine, Alanine, Tyrosine, Stop, Histidine, Glutamine, Asparagine, Lysine, Aspartic acid, Glutamic acid, Cysteine, Arginine, and Glycine.
    c. Stop, and Serine (if couple with wobble in the first position)
    d. Arginine, Leucine, and Serine (if coupled with wobble in the second position)
    e. Serine (though wobble in the first and second positions have to occur together)
    f. If this hypothesis is correct, the 3rd position likely has the most mutations since that is the one best 'protected' by wobble.
7.

Gene 1:

| | |
|---|---|
| mRNA strand | AUG UCA CAU UUU GAC UGG CGU CUC GAA UAG |
| amino acid chain | MET, SER, HIS, PHE, ASP, TRP, ARG, LEU, GLU, STOP |

Gene 2:

| | |
|---|---|
| mRNA strand | AUG AAA CCC GGG UUU AUU CUU AAC AGA CGU CCG AUA UAA |
| amino acid chain | MET, LYS, PRO, GLY, PHE, ISO, LEU, ASN, ARG, ARG, PRO, ISO, STOP |

Gene 3:

| | |
|---|---|
| mRNA strand | AUG GUU AUU GAG ACU UUG CAG CGA CGU AGA UAC AAU UGA |
| amino acid chain | MET, VAL, ISO, GLU, THR, LEU, GLN, ARG, ARG, ARG, TYR, ASN, STOP |

8.
    a. Only the sixth amino acid is different. Trp is coded for by the codon UGG, so the mutation must have been a substitution of G for U in the 2nd position of the 6th codon.
    b. Substitution mutation: the correct codon is GAG, a substitution of U for G in the first position would create this amino acid chain
    Frameshift: Since UGA is also a stop codon, an insertion of a U between the third and fourth codons would also produce this effect

9. Answer is B and E
10. Answer is B
11.
    a. Codominance
    b. Incomplete dominance
    c. Simple dominance
    d. Pleiotropy and simple dominance
    e. Environmental effects on phenotype
    f. Polygenic inheritance
    g. Epistasis

12. Genotype is the genetic makeup of an organism (the alleles it carries), phenotype is the traits it exhibits (its physical appearance). The phenotype is determined by the genotype and by the environment
13. Heterozygous means having two different alleles for a given gene, homozygous means two copies of the same allele
    a. To exhibit the recessive phenotype, the white flower must be homozygous recessive
    b. The dominant phenotype is exhibited by both the homozygous dominant genotype and the heterozygous genotype. Given only this information, we cannot be sure of the genotype of this plant.

14. Genetic diversity is important for maintaining a healthy population and responding to new selective pressures (we will get into more detail on this in the next couple weeks)
    a. 'shufffles' whole genomes from two different individuals
    b. 'shuffles' sets of chromosomes and allows an organism to make millions of genetically unique gametes
    c. 'shuffles' alleles on a given chromosome, creates new chromosome combinations different than what the organism inherited
15.

a.

| | R | R |
|---|---|---|
| r | Rr (white) | Rr (white) |
| r | Rr (white) | Rr (white) |

b.

| | R | r |
|---|---|---|
| r | Rr (white) | rr (yellow) |
| r | Rr (white) | rr (yellow) |

c.

| | R | r |
|---|---|---|
| R | RR (white) | Rr (white) |
| r | Rr (white) | rr (yellow) |

16.

a.

| | T | T |
|---|---|---|
| t | Tt (tall) | Tt (tall) |
| t | Tt (tall) | Tt (tall) |

b.

| | G | g |
|---|---|---|
| g | Gg (green) | gg (yellow) |
| g | Gg (green) | gg (yellow) |

c.

| | B | B |
|---|---|---|
| b | Bb (brown) | Bb (brown) |
| b | Bb (brown) | Bb (brown) |

d.

| | R | r |
|---|---|---|
| R | RR (white) | Rr (white) |
| r | Rr (white) | rr (brown) |

17. 25%

| | A | a |
|---|---|---|
| A | AA (normal) | Aa (normal) |
| a | Aa (normal) | aa (albino) |

18. From your prediction, you believed this trait was complete/simple dominant and that your red-flowered plant was heterozygous. The results of your cross indicate that the trait is actually intermediate/incomplete dominant and your plants were both homozygous.

19. The fact that the entire F1 generation is smooth, and the F2 generation is ~3:1 smooth indicates that smooth is the dominant trait and that the initial parents were homozygous dominant and homozygous recessive

20. This pattern of inheritance (F1 all brown, F2 3 brown:1 silverblue) indicates that brown is the dominant phenotype and the original parents were homozygous dominant and homozygous recessive. The F1 offspring are all heterozygous and we would expect the F2s to be 1 BB:2 Bb:1 bb. We would expect 50% of the F2s to be homozygous (25% homozygous dominant and 25% homozygous recessive)

| | B | B |
|---|---|---|
| b | Bb (brown) | Bb (brown) |
| b | Bb (brown) | Bb (brown) |

F1 offspring

| | B | b |
|---|---|---|
| B | BB (brown) | Bb (brown) |
| b | Bb (brown) | bb (silverblue) |

F2 offspring

21.
   a. Parent genotypes are TTpp and ttPP, all F1 offspring will be TtPp, tall and purple
   b. Parent genotypes are TtPp and TTpp; Offspring will be 50% tall purple, 50% tall white

| | TP | Tp | tP | tp |
|---|---|---|---|---|
| Tp | TTPp (Tall, purple) | TTpp (Tall, white) | TtPp (tall, purple) | Ttpp (tall, white) |

22.
   a. Smooth green seeds are produced by the following genotypes: ggWW and ggWw
      These two genotypes can be produced in three different sperm/egg combinations:
      P(gW sperm) and P(gW egg) = 0.25 x 0.25 = 0.0625
      P(gW sperm) and P(gw egg) = 0.25 x 0.25 = 0.0625
      P(gw sperm) and P(gW egg) = 0.25 x 0.25 = 0.0625
      0.0625 + 0.0625 + 0.0625 = **0.1875**
   b. Wrinkled yellow seeds are produced by the following genotypes: GGww and Ggww
      These two genotypes can be produced in three different sperm/egg combinations:
      P(Gw sperm) and P(Gw egg) = 0.25 x 0.25 = 0.0625
      P(Gw sperm) and P(gw egg) = 0.25 x 0.25 = 0.0625
      P(gw sperm) and P(Gw egg) = 0.25 x 0.25 = 0.0625
      0.0625 + 0.0625 + 0.0625 = **0.1875**

   c. Now we are looking at just one genotype: GgWw, which can be produced by four different sperm/egg combinations:
      P(GW sperm) and P(gw egg) = 0.25 x 0.25 = 0.0625
      P(Gw sperm) and P(gW egg) = 0.25 x 0.25 = 0.0625
      P(gW sperm) and P(Gw egg) = 0.25 x 0.25 = 0.0625
      P(gw sperm) and P(GW egg) = 0.25 x 0.25 = 0.0625
      0.0625 + 0.0625 + 0.0625 + 0.0625 = **0.25**

23.
   a. BB, Bb, and bb
   b. BB and Bb are blue, bb is red
   c. BB is blue, bb is red, and Bb is blue and red stripes or splotches (both homozygous phenotypes are expressed)
   d. BB is blue, bb is red, and Bb is purple (homozygous phenotypes blended)

24.

a. Blue
b. White
c. Blue
d. Magenta
e. To solve this problem, start with what is known about the genotypes of each plant given its flower color and then fill in the other alleles based on the inheritance pattern.
The known information just from the flower color is __ff x bbF_ = B_F_
Given that the magenta parent is bb, the offspring must be Bb and the white parent must be BB (if the white parent was Bb, some of the offspring would be magenta).
Given that the white parent is ff, the offspring must be Ff and the magenta parent must be FF (if the magenta parent was Ff, some of the offspring would be white).
Consequently, we can fill in all the alleles: BBff x bbFF = BbFf

25.

| Genotype | AA+/+ | AB+/- | AB+/+ | OO-/- | AO+/- | BO-/- | BB+/- | AB-/- |
|---|---|---|---|---|---|---|---|---|
| Phenotype | A-pos | AB-pos | AB-pos | O-neg | A-pos | B-neg | B-pos | AB-neg |

26. To answer this question, list all the possible genotypes for the parents and the baby and then check if any combination of parental genotypes could produce any of the offspring genotypes.

| Father | Mother | Baby |
|---|---|---|
| AA+/+ | BB+/+ | OO-/- |
| AO+/+ | BO+/+ | |
| AA+/- | BB+/- | |
| AO+/- | BO+/- | |

If both parents are heterozygous for both traits (AO+/- and BO+/-), then they could have an OO-/- baby. There is no evidence this baby was switched

Sex-Linked Inheritance

27.

| Genotype | Sex | Disease Status (healthy, carrier, or hemophiliac) |
|---|---|---|
| $X^HX^H$ | Female | Healthy |
| $X^HY$ | Male | Healthy |
| $X^hX^h$ | Female | Hemophiliac |
| $X^hY$ | Male | Hemophiliac |
| $X^HX^h$ | Female | Carrier |

28. Overall, 50% of offspring will have hemophilia. In this cross, the probability is the same in both sexes

| | $X^h$ | Y |
|---|---|---|
| $X^H$ | $X^HX^h$ (F, carrier) | $X^HY$ (M, healthy) |
| $X^h$ | $X^hX^h$ (F, hemo) | $X^hY$ (M, hemo) |

29. 25% of children will have hemophilia, all will be male

Daughters will be 50% healthy non-carriers and 50% carriers

| | $X^H$ | $Y$ |
|---|---|---|
| $X^H$ | $X^HX^H$ (F, healthy) | $X^HY$ (M, healthy) |
| $X^h$ | $X^HX^h$ (F, carrier) | $X^hY$ (M, hemo) |

30. All male offspring will have hemophilia (expected 50% of total offspring). All female offspring will be carriers.

| | $X^H$ | $Y$ |
|---|---|---|
| $X^h$ | $X^HX^h$ (F, carrier) | $X^hY$ (M, hemo) |
| $X^h$ | $X^HX^h$ (F, carrier) | $X^hY$ (M, hemo) |

31. Since men only have one X chromosome, a single disease allele will cause them to exhibit the disease. With two X chromosomes, women must inherit two disease alleles in order to exhibit the disease. Since these alleles are often somewhat rare, it is less likely for a woman to inherit two disease alleles than for a man to inherit one.

32. None. A man inherits a Y chromosome from his father. If he has the disease, he inherited it from his mother through the X chromosome he inherited from her.

33.

a. $X^RX^r$ Female, Red $X^RY$ Male, Red

$X^RX^R$ Female Red $X^rY$ Male, White

b. white eyed, male $X^rY$ red eyed female $X^RX^R$, $X^RX^r$

white eyed, female $X^rX^r$ red eyed, male $X^RY$

| | $X^R$ | $Y$ |
|---|---|---|
| $X^r$ | $X^RX^r$ (F, Red) | $X^rY$ (M, white) |
| $X^r$ | $X^RX^r$ (F, red) | $X^rY$ (M, white) |

34.

a. None, because they inherit a white allele from the mother
b. None, because they inherit a red allele from the father

<u>Pedigrees</u>

35.

a. We cannot be certain, they could be either. Given that the parents are both heterozygous, and IV-2 does have hitchhiker's thumb, they have a 33% chance of being HH and a 66% chance of being Hh.
b. The double line represents inbreeding. Persons III-1 and III-2 are first cousins
c. Related individuals are more likely to carry the same recessive alleles. That means their offspring are more likely to inherit the recessive allele from both parents and express the recessive phenotype. This is one of the many reasons people should not mate with close relatives.

36. It is possible these genes are linked. If genes contributing to tameness are on the same chromosomes as genes relating to floppy ears, curly tails, etc, then breeding for tameness will also increase these other traits in the population. It is also possible that the genes for tameness are pleiotropic and also impact these physical traits directly.

37.

a. Recessive. Carla has the condition and neither of her parents do (they must be heterozygous and she inherited the recessive alleles)
b. All purple individuals are aa. Ann and Michael must be Aa in order to have aa offspring. The same is true for George. Sam and Ann must be Aa since they inherited an a allele from one parent (Arlene) and do not exhibit the trait. The same is true for Daniel and Alan. Sandra, Tina, and Christopher are A_, but we cannot determine the second allele with certainty.
Note: students will think we can be certain Sandra is AA given that both her children carry an A allele. Two offspring is far too small a sample size to be certain of this. Even if she is Aa, there is a 25% chance that her two offspring both inherit her A allele

Linkage Mapping

38. Yes. The crosses first determine if G and F are linked and then if G and C are linked. If these genes were unlinked, we would expect the resulting offspring ratios to be even. The offspring ratios are very uneven in both crosses, meaning that genes G and F are linked and genes G and C are linked. This then means that all three genes are on the same chromosome, indicating that F and C are also linked.

39.

a. Genes L and M are linked and genes G and B are linked, which you can tell because their gamete frequency is uneven. If genes are unlinked, we expect their distribution into gametes to be roughly even (like genes L and G, which are unlinked). Given this information, we can also determine that genes M and B are unlinked (because M is on the same chromosome as L, and G is on the same chromosome as B and we know that L and G are on different chromosomes)

b. The closer together two genes are on a chromosome, the less frequently we will see their alleles be separated by recombination. Alleles from genes G and B are inherited together more frequently (their gamete frequencies are more uneven) than alleles from genes L and M, meaning that genes G and B are closer together on the chromosome than genes L and M.

c. There are multiple gamete combinations that would produce a *GgBb* offspring, however not all of them are possible in this cross. Let's call Parent 1 the *GgBb* parent and Parent 2 is the *ggbb* parent. Parent 1 can produce all possible combinations of gametes, while Parent 2 only produces *gb* gametes. Additionally, the gametes that Parent 1 produces are not in the even frequencies that Mendel's Laws predict, but instead are produced in the frequencies provided in the table.

   Since Parent 2 can only produce *gb* gametes, there is only one possible inheritance pattern that would produce *GgBb* offspring:
   f(*GgBb*) = f(*GB*) x f(*gb*) = 0.47 x 1 = 0.47

d. Normal antennae and normal eyes are produced by four different genotypes (*GgBb, GGBb, GgBB,* and *GGBB*); however, not all of these genotypes are possible given the genotypes of our parents (which are *GgBb* and *ggbb*). The only genotype that produces normal antennae and normal eyes from this cross is *GgBb*, so our answer is the same as to part c above: 0.47